

Detailed breakdown of **all the common components that we will see under the SQL Server 2019 Setup Feature Selection**, grouped into **Instance (instance-aware) Features** and **Shared Features**, with explanations, purpose, install-time notes, typical configuration considerations, and best-practice tips.

Used Microsoft's installation docs as the source for what is instance-aware vs shared and for install behavior. ([Microsoft Learn](#))

Instance-aware Features (features that belong to a specific SQL Server instance)

These features are installed *for a particular instance* and versioned with that instance. They can be added to an existing instance (except for some clustered cases). ([Microsoft Learn](#))

1. Database Engine Services

- **What it is:** Core service that stores, processes and secures the relational databases.
- **Includes:** Engine, system databases (master, model, msdb, tempdb), SQL Server Agent (scheduling), basic instance configuration (authentication mode, collations, tempdb layout).
- **When to choose:** Required for any SQL Server instance that will host databases.
- **Notes / best practice:** Configure service accounts, TempDB file count & sizes at install time (or plan to change immediately afterwards). Grant Perform Volume Maintenance Task for instant file initialization if you want faster data file allocation. ([Microsoft Learn](#))

2. Analysis Services (AS)

- **What it is:** OLAP (Multidimensional) and Tabular analytical engine.
- **Modes:** Tabular or Multidimensional (you choose one during install).
- **Use case:** Data warehouses, cubes, Power BI on-prem scenarios, BI semantic models.
- **Notes:** Choose correct mode; AS is an independent service/instance and requires separate ports, service account, and storage planning.

3. Reporting Services (SSRS) — Native / (note)

- **What it is:** Reporting engine for paginated reports.
- **Important note for 2017/2019+:** Historically SSRS could be selected in the SQL setup. In recent releases Microsoft decoupled SSRS and it's often installed/updated separately (Power BI Report Server / SSRS installer). Check your installer/version. If shown in your feature list it will appear as Reporting Services. ([Microsoft Learn](#))
- **Notes:** SSRS requires configuration after install (report server DB, service account, URL bindings).

4. Integration Services (SSIS)

- **What it is:** ETL engine for package execution (SSIS).
- **Use case:** Data movement, transformations, ETL workflows.
- **Notes:** Install the service and optionally install client components for development. SSIS package catalog (SSISDB) requires a database that you create/configure after install.

5. Machine Learning Services (In-Database)

- **What it is:** Run R and/or Python code inside SQL Server (in-database analytics).
- **Use case:** Scoring models, data science workloads where you want models close to the data.
- **Notes:** Can require additional packages; pick R, Python or both. Requires enabling external script execution and correct service account permissions.

6. PolyBase Query Service for External Data

- **What it is:** Enables querying external data sources (Hadoop, Oracle, Teradata, MongoDB, etc.) via T-SQL (data virtualization).
- **Use case:** Data virtualization and big data integration.
- **Notes:** PolyBase requires Java (SQL Server 2019 can install Zulu OpenJRE) and may require additional configuration (scale-out groups, ports). Certain PolyBase scenarios add extra Windows services or dependencies. ([Microsoft Learn](#))

7. Full-Text and Semantic Extractions for Search (Full-Text Search)

- **What it is:** Full-text indexing and search capability for text columns; semantic search adds entity extraction and similarity.
- **Use case:** Rich text search in apps (CONTAINS, FREETEXT).
- **Notes:** Can be added to an existing instance if missing; full-text catalogs live in the database but service components are instance-level.

8. Replication

- **What it is:** SQL Server replication components (snapshot, transactional, merge).
- **Use case:** Offloading reporting, near real-time distribution, heterogeneous sync scenarios.
- **Notes:** After install you still must configure publishers, distributors and subscribers. Replication agents run under specified accounts.

9. Data Quality Services (DQS)

- **What it is:** Data cleansing, matching, knowledge base driven de-duplication feature (less common).
- **Use case:** Data quality / cleansing workflows.
- **Notes:** DQS components sometimes require additional configuration and a DQS database.

10. Distributed Replay (controller/client)

- **What it is:** Replay captured workloads for testing (performance regression testing).
- **Use case:** Test upgrades, validate performance.
- **Notes:** Components include controller and client and require separate machines in larger tests.

11. (Other instance-aware components you may see)

- **Java Language Extensions / Java Install Location (2019+):** Java runtime selection for PolyBase or language extensions. ([Microsoft Learn](#))
- **Big Data Cluster components** (if using big data cluster images)—these are special and often separate installers or containerized.

Shared Features (components installed once and shared across instances on the machine)

Shared features are installed to shared directories and are not tied to a single SQL Server instance. They are used by multiple instances and tools. You select or change the shared feature install path on the Feature Selection page. ([Microsoft Learn](#))

1. Client Tools Connectivity

- **What it is:** Network libraries, client connectivity components, SQL Native Client pieces required for applications and tools to connect to SQL Server.
- **Use case:** Ensures ODBC/OLE DB/ADO.NET client drivers and connection libraries are available.

- **Notes:** Important on application servers and developer boxes.
- 2. **Client Tools Backwards Compatibility**
 - **What it is:** Older client components required for legacy apps.
 - **Use case:** If older applications need legacy drivers or tools.
- 3. **Management Tools — Basic & Complete**
 - **Management Tools — Basic:** Minimal management utilities (sqlcmd, basic tools).
 - **Management Tools — Complete:** Includes more advanced tools; historically this was where SQL Server Management Studio (SSMS) components were referenced. **Note:** SSMS is now a separate download and the installer may not include the full SSMS GUI; still, Management Tools entries appear for feature completeness. ([Microsoft Learn](#))
 - **Use case:** DBA machines, server admin tasks.
 - **Notes:** After install, you'll typically download the latest SSMS separately.
- 4. **Connectivity Components / SDK / Client Tools SDK**
 - **What it is:** SDKs, header files, tools for building client apps that talk to SQL Server (less common on production servers).
 - **Use case:** Development and integration work.
- 5. **Documentation Components**
 - **What it is:** Local help and documentation files (optional).
 - **Use case:** Offline help for environments without internet access.
- 6. **Integration Services (sometimes appears as shared)**
 - **What it is:** In some installer views the SSIS service & runtime components appear under shared features as well—because the runtime and client design components can be shared across instances.
 - **Notes:** SSIS catalog (SSISDB) is created per instance (in a database), but the runtime binaries are shared.
- 7. **Reporting Services Add-ons for SharePoint / SharePoint Integration**
 - **What it is:** If you integrate with SharePoint, there are shared components and add-ins used by multiple instances.
 - **Notes:** Modern SSRS + SharePoint scenarios vary by SQL/SharePoint version.
- 8. **Master Data Services (MDS) / Power BI Report Server (where applicable)**
 - **What it is:** These appear in some installers or documentation as add-on/server products. They are typically set up separately but may show in feature lists depending on installer package.

Practical install-time notes & best practices (short checklist)

- **Decide instance vs shared location:** Use a dedicated drive for *Instance root directory* and a different drive for *Shared feature directory* (x86) — do **not** install shared features on the OS drive if possible. ([Microsoft Learn](#))
- **Service accounts:** Use dedicated domain accounts for production service accounts (DB engine, SSAS, SSIS) — do not use LocalSystem if network access or Kerberos is required.
- **TempDB sizing:** Configure TempDB files and sizes at install or immediately after; it's much easier than resizing on the fly under load. ([Microsoft Learn](#))
- **Feature dependencies:** Some features (PolyBase, Machine Learning, SSRS) add dependencies (Java, R/Python runtimes, separate installers). PolyBase will prompt for Java/JRE location in 2019+. ([Microsoft Learn](#))

- **Failover cluster limitations:** You cannot add certain features to an existing failover cluster instance (e.g., PolyBase). If you need them, plan them before clustering. ([Microsoft Learn](#))
- **Post-install:** Apply the latest CUs/service packs so all installed components are at the same patch level (especially if you add features later). ([Microsoft Learn](#))

Quick mapping table (feature name → short purpose)

Feature	Short purpose
Database Engine Services	Core DB engine (stores/serves data)
Analysis Services	OLAP/Tabular analytics engine
Integration Services	ETL / package runtime (SSIS)
Reporting Services (SSRS)	Paginated reporting engine (may be separate installer)
Full-Text & Semantic Search	Text indexing & advanced search
Replication	Data distribution (snapshot/txn/merge)
Machine Learning Services	Run R/Python in-DB
PolyBase Query Service	Query external data sources (data virtualization)
Data Quality Services	Data cleansing & matching
Distributed Replay	Workload replay for testing
Management Tools (Basic/Complete)	Admin tools and client utilities
Client Tools / Connectivity / SDK	Drivers & client libraries